조선대학교
CHOSUN UNIVERSITY

실습 결과 제출

기계학습
- 실습환경 구축하기 (feat. Docker) -

2026. 03. 09

20244792 AI·SW학부(인공지능공학전공)

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about ‘Docker’

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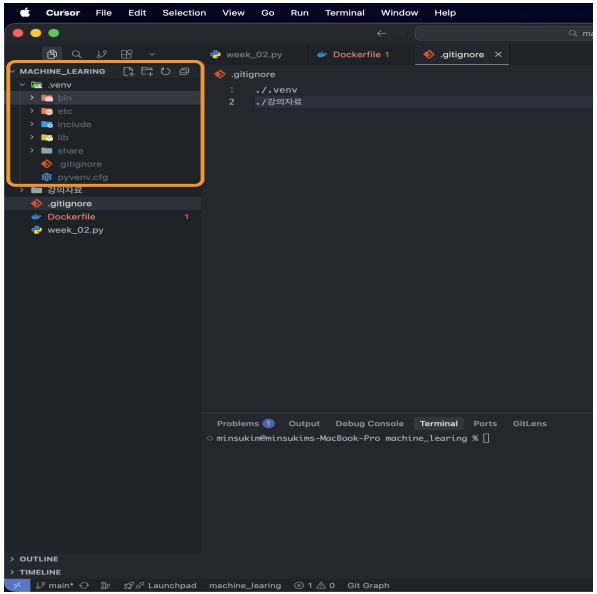
- 개인 컴퓨터 별로 개발 및 실행 환경을 통일 시키기 어렵다
- 프로그램에 필요한 Dependency등을 가상 컴퓨터에 설치해 통일
- container/[python3.12.x, jupyterlab, …] 이렇게 layer가 있다

(본인은 cursor IDE를 사용하였습니다.)

시행착오: 로컬 가상환경과 Docker 가상환경의 차이

1. python3 -m venv .venv
2. source .venv/bin/activate
3. pip install jupyterlab numpy pandas matplotlib scikit-learn seaborn



여기서 문제가 된 것은 venv는 내 로컬 python으로 만드는 로컬 전용 가상환경이라는 것이다.

우리 수업의 목표는 Docker로 환경 세팅 후 이미지를 만들고, 실행 및 마운트 하는 것이었다.

Docker로 처음부터 시작하기

Dockerfile을 먼저 작성해 base image (pytorch/pytorch) 설정
requirements.txt에 dependency 정리 (pip freeze > requirements.txt)

pip freeze: dependency 출력

> or >>: redirection (linux 명령어)

(! important) m1은 GPU 가속 CUDA 지원 X
python-slim + torch

Dockerfile M | requirements.txt M | .gitignore

Dockerfile > ...

You, 20 seconds ago | 1 author (You)

```dockerfile
FROM python:3.11-slim

WORKDIR /app

ENV PYTHONDONTWRITEBYTECODE=1 \
    PYTHONUNBUFFERED=1 \
    PIP_NO_CACHE_DIR=1

RUN apt-get update && apt-get install -y --no-install-recommends \
    build-essential \
    git \
    curl \
    && rm -rf /var/lib/apt/lists/*

COPY requirements.txt .

RUN python -m pip install --upgrade pip && \
    python -m pip install -r requirements.txt

COPY . .

EXPOSE 8888        You, 21 seconds ago • Uncommitted changes

CMD ["jupyter", "lab", "--ip=0.0.0.0", "--port=8888", "--no-browser", "--allow-root"]
```

Dockerfile M | requirements.txt M | .gitignore

requirements.txt

```requirements.txt
torch==2.10.0
torchvision==0.25.0
torchaudio==2.10.0
jupyterlab==4.2.5
numpy==1.26.4
pandas==2.2.3
matplotlib==3.8.4
scikit-learn==1.5.2
seaborn==0.13.2
```

⌘L to chat, ⌘K to generate

Docker로 빌드하기

```docker build --platform linux/arm64 -t univ_chosun_ml .```
dockerfile image 생성

(```docker images```로 확인 가능)

```
Problems   Output   Debug Console   Terminal   Ports   GitLens

minsukim@minsukims-MacBook-Pro machine_learing %
minsukim@minsukims-MacBook-Pro machine_learing %
minsukim@minsukims-MacBook-Pro machine_learing % docker build --platform linux/arm64 -t univ_chosun_ml .
[+] Building 1.1s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 445B
 => [internal] load metadata for docker.io/library/python:3.11-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/6] FROM docker.io/library/python:3.11-slim@sha256:d6e4d224f70f9e0172a06a3a2eba2f768eb146811a349278b38fff3a36463b47
 => => resolve docker.io/library/python:3.11-slim@sha256:d6e4d224f70f9e0172a06a3a2eba2f768eb146811a349278b38fff3a36463b47
 => [internal] load build context
 => => transferring context: 2.46MB
 => CACHED [2/6] WORKDIR /app
 => CACHED [3/6] RUN apt-get update && apt-get install -y --no-install-recommends     build-essential     git     curl
 => CACHED [4/6] COPY requirements.txt .
 => CACHED [5/6] RUN python -m pip install --upgrade pip &&     python -m pip install -r requirements.txt
 => CACHED [6/6] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:a3720eccbf34167f9f6a53ba9c13deaf8a12260c2cb7d3bbce50c4b0fa6f399c
 => => exporting config sha256:d6b08a444213a3e11be00045440ae6d67d6f5630063adaac1ebe5f6ffd847223
 => => exporting attestation manifest sha256:26b405d22c9fb38388c07eb7c8beb53dc879a5fc494afa323d3cb09370226258
 => => exporting manifest list sha256:b5c932d37860766642eac41299a1c7f6300e564e00ba5c3bbf394cc71f3c4ad0
 => => naming to docker.io/library/univ_chosun_ml:latest
 => => unpacking to docker.io/library/univ_chosun_ml:latest

⌘K to generate command
machine_learing   ⊗ 0 ⚠ 0   Git Graph   -o- You, now   Cursor Tab   -o- Not Committed Yet   Ln 6, Col 14
```

docker build -t
<image_name> <target>
('.' = all)

(! important)
--platform linux/arm64
로 cpu 아키텍처 명시 (m1)
```

Docker image 실행

docker run --rm -it \ -p 8888:8888 \ -v "$(pwd)":/app \ -v "$HOME/.jupyter:/root/.jupyter:ro" \ -w /app \ univ_chosun_ml

-v $(pwd) => 로컬 수정 반영 | -v ~/.jupyter => *_config.json 반영

```
Problems   Output   Debug Console   Terminal   Ports   GitLens

minsukim@minsukims-MacBook-Pro machine_learing % docker run --rm -it -p 8888:8888 -v "$(pwd)":/app -w /app univ_chosun_ml
[I 2026-03-09 12:31:09.728 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-03-09 12:31:09.730 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-03-09 12:31:09.731 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-03-09 12:31:09.733 ServerApp] Writing Jupyter server cookie secret to /root/.local/share/jupyter/runtime/jupyter_cookie_secret
[I 2026-03-09 12:31:09.939 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-03-09 12:31:09.946 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-03-09 12:31:09.947 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-03-09 12:31:09.947 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-03-09 12:31:09.948 LabApp] JupyterLab extension loaded from /usr/local/lib/python3.11/site-packages/jupyterlab
[I 2026-03-09 12:31:09.948 LabApp] JupyterLab application directory is /usr/local/share/jupyter/lab
[I 2026-03-09 12:31:09.948 LabApp] Extension Manager is 'pypi'.
[I 2026-03-09 12:31:09.955 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-03-09 12:31:09.956 ServerApp] Serving notebooks from local directory: /app
[I 2026-03-09 12:31:09.956 ServerApp] Jupyter Server 2.17.0 is running at:
[I 2026-03-09 12:31:09.956 ServerApp] http://1e4ddc7cf265:8888/lab?token=dc6f586dcb23afc6b0f3f20b830e5e6d60c67fdc19355bdd
[I 2026-03-09 12:31:09.956 ServerApp]     http://127.0.0.1:8888/lab?token=dc6f586dcb23afc6b0f3f20b830e5e6d60c67fdc19355bdd
[I 2026-03-09 12:31:09.956 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-03-09 12:31:09.957 ServerApp]

    To access the server, open this file in a browser:
        file:///root/.local/share/jupyter/runtime/jpserver-1-open.html
    Or copy and paste one of these URLs:
        http://1e4ddc7cf265:8888/lab?token=dc6f586dcb23afc6b0f3f20b830e5e6d60c67fdc19355bdd
        http://127.0.0.1:8888/lab?token=dc6f586dcb23afc6b0f3f20b830e5e6d60c67fdc19355bdd
[I 2026-03-09 12:31:09.965 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-type
script-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-s
erver, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver
-bin, yaml-language-server

⌘K to generate command

machine_learing  ⊗ 0 ⚠ 0  Git Graph   -o- You, 13 seconds ago  Cursor Tab  Ln 6, Col 14  Spaces: 4  UTF-8  LF  pip requirements
```

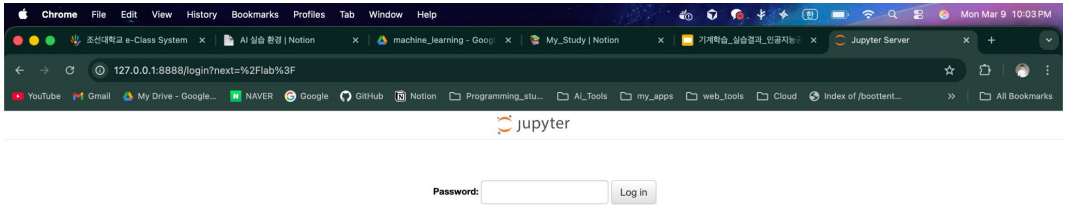
-it: 터미널 실행
--rm: 종료시 컨테이너 삭제
-v: 도커 볼륨 마운트
-p: 포트 열기
sh: shell 실행

-w: 컨테이너가 실행될 때 현재 디렉토리를 지정한 경로로 설정

마운트하면 컨테이너가 로컬 폴더를 직접 읽음

jupyter server *_config.json (*_config.py)

password만 받도록 했음
(token도 입력하길 원하면 *_config.py에서 c.ServerApp_token = <str> 채우기)



QnA (with codex)

home/.jupyter랑 왜 굳이 root: ro로 연결해?

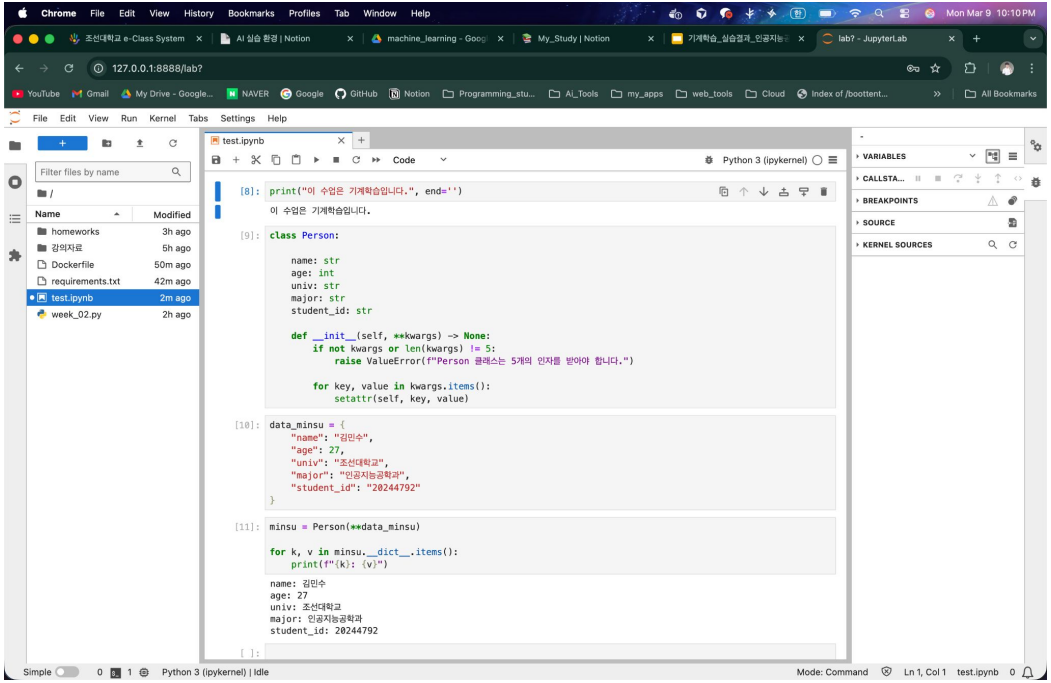
/root/.jupyter:
- 컨테이너 프로세스가 root 사용자로 실행 중이라 Jupyter가 설정을 ~/.jupyter = /root/.jupyter 에서 찾습니다.

$HOME/.jupyter:/root/.jupyter:
- 호스트에서 관리하는 설정(토큰/비번 정책)을 컨테이너가 그대로 쓰게 하려는 목적입니다.

:ro:
- 컨테이너가 호스트 설정 파일을 덮어쓰지 못하게 하는 안전장치(읽기 전용)입니다.

jupyter server test

내 정보 클래스 만들고 출력 완료



마지막 정리:

1. 로컬과 도커 헷갈림
2. 도커 빌드 시 arm 오류
3. 도커 실행 시 mount 해 로컬 및 config 반영